

Week 1: Introduction (I)

PLSC 309 Quantitative Political Analysis

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Department of Political Science
Penn State

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Welcome!

Part 1. Introduction

Introduction

- About Myself
- About You
- About This Course

- I'm Yang Yang.
- ... 5th year PhD student in Political Science and Social Data Analytics.
- My research focuses on International Relations and Political Methodology.
- You can just call me Yang.

About My Background

Before grad school in the U.S.,

- Intern journalist in Bangkok, Thailand
- Policy analyst in Shanghai, China.



- Substantive: International relations, international conflict, foreign policy, diplomacy
- Methodological: Causal inference, machine learning, natural language processing, . . .

DeepLearningCausal: R Package for Estimating Treatment Effects Using Deep Neural Networks and Ensemble Learning

by Nguyen Khoi Huynh, Yang Yang, and Bumba Mukherjee

Abstract Conventional machine learning (ML) methods for causal inference are widely used for estimating causal effects and uncovering heterogeneous treatment effects. Yet, deep learning, which leverages deep neural networks to address selection bias and unobserved confounders, is under-utilized for estimating sample and population average treatment effects. Further, conventional ML methods are employed to estimate treatment effects under the assumption that all units in the sample comply with the treatment, which is often implausible. To address these methodological challenges, this article introduces our DeepLearningCausal R-package that enables users to employ deep neural networks to estimate conditional average treatment effects (CATEs) from samples via meta-learners and population average treatment effects on the treated (PATT) in settings with treatment noncompliance. DeepLearningCausal also includes functions to compute and illustrate conformal prediction (CP) intervals for the estimated meta-learner individual treatment effects (ITEs), visualize heterogeneous treatment effects, and estimate CATEs and PATT by using weighted ensemble learning.

The Limitations of Using Forced Choice in Electoral Conjoint Experiments *

Giancarlo Visconti[†]

Yang Yang[‡]

Abstract

Electoral conjoint experiments typically require respondents to choose between two candidates, assuming full turnout and strict preference orderings, even though real-world voters may abstain or cast protest votes. We show that excluding these options is not trivial. In our preregistered experiment, more than half of respondents abstained or cast protest votes at least once when such options were available. While most AMCE estimates remain stable, a subset shifts in direction, magnitude, or significance—changes that are unpredictable *ex ante* and can mislead inferences about voter preferences. Marginal mean estimates diverge even sharply across designs. These shifts are systematic, shaped by respondent-level traits (e.g., turnout history and propensity) and task-level features such as rating differentials and contest appeal. Our findings show that forced-choice designs impose strong behavioral assumptions and risk obscuring meaningful variation in electoral behavior. Providing explicit opt-out options enhances validity and better captures the full range of voter preferences.

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- Email address: yang.yang@psu.edu

About You

- Your name
- Your major and year
- Why are you taking this course?
- Any prior exposure to politics, such as:
 - Any political science courses you have taken
 - Experience with political participation (e.g., voting, volunteering, internships)
- Any statistical, quantitative, or programming skills you have (e.g., Excel, R, Python, SPSS, STATA)

Course Syllabus:

[PLSC 309 Fall 2026 Syllabus](#) (Posted on Canvas)

- What is this course about?
- What are our goals?
- How will we learn?
- What materials do we need?
- How is the course organized?
- How will you be evaluated?
- What are the course policies?
- Questions and Discussion

What Is This Course About?

This is not an econometrics or statistics course.

The emphasis is on developing intuition and learning how to use quantitative methods appropriately in political (social) science:

- How to design quantitative research
- How to choose and apply appropriate methods
- How to interpret and evaluate statistical results
- How to draw appropriate conclusions from empirical evidence

No prior background in statistics or programming is required.

What Are Our Goals?

By the end of this course, you will be able to:

- critically read and understand empirical political (social) science research.
- analyze and visualize data using R.
- design and carry out an independent quantitative research project.
- feel empowered working with data.

- The course will challenge you, but it is designed to be manageable.
- My primary goal is to show that scientific study of politics (social sciences) is both rigorous and interesting.

How Will We Learn?

- Two in-person meetings each week:
 - Tue and Thu (1:35-2:50 pm)
- Each meeting includes:
 - Lecture: concepts and methods
 - Lab: hands-on practice in R, as appropriate
- Bring your laptop to every class for in-class exercises and R labs.
- Install R and RStudio before the next class. (instructions available on page 3 of the syllabus)

What Materials Do We Need?

- Think of the **lecture slides** as primary reading
- **Required textbook**
 - Kellstedt & Whitten (KW), *The Fundamentals of Political Science Research* (2nd or 3rd ed.)
- **Other reading**
 - Posted on Canvas

How the Course Is Organized

Weeks	Modules
Week 1	Understanding Quantitative Political Research
Week 2	Variables, Measurement, and Data
Weeks 3–4	Fundamentals of Probability and Statistics
Weeks 5–9	Linear Regression
Week 10	Data Wrangling and Visualization
Weeks 11–12	Advanced Regression Models
Week 13	Intro to Causal Inference
Week 14	Intro to Computational Social Science
Weeks 15–16	Final Project

How Will You Be Evaluated?

Full details on Pages 4-6 of the Syllabus

- In-class participation (20%)
 - Attendance 10%
 - Active participation 10%
- Problem sets (35%)
- Midterm exam (15%)
- Final research project (30%)
 - Four checkpoints throughout the semester 20% (5% each)
 - 3-min presentation 5%
 - Written paper and R script 5%

What Are the Course Policies?

- AI may be used for coding assistance, but not to generate graded interpretations or written answers.
- All submitted work must comply with academic integrity requirements.
- Treat others with respect and help maintain an inclusive classroom environment.

Questions?

Any questions about this class:

- Course requirements?
- Assignments and grading?
- The final project?
- Course policies?

Read the [syllabus](#) for more details.

Anonymous Course Feedback

In addition to Penn State's end-of-semester SEEQ course evaluation, I have created an anonymous feedback form for this course.

- Share feedback at any time during the semester.
- Tell me what is working well and what could be improved.
- All responses are completely anonymous.
- Your feedback will help us create a better learning experience and environment together.

[Anonymous Course Feedback - Fill out form](#)

Part 2. The Scientific Study of Politics

What Do Political Scientists Do?

Many people work on political issues, but they approach them in fundamentally different ways.

- Journalists
- Lawyers
- Political consultants
- Activists
- Political scientists

How are political scientists different?

What Do Political Scientists Do?

Profession	Primary Question
Journalists	What happened?
Lawyers	Is this legal or constitutional?
Political consultants	How can my candidate win?
Activists	What should government do?
Political scientists	Why did these things happen? What explains such political behaviors?

Social Science Is a Science

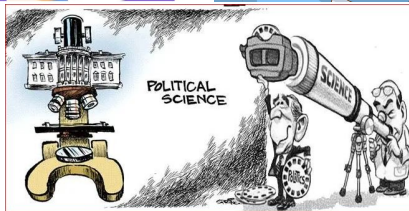
Social scientists study **human behavior**, including:

- Preferences
- Decision-making
- Interactions with others
- Responses to institutions and environments

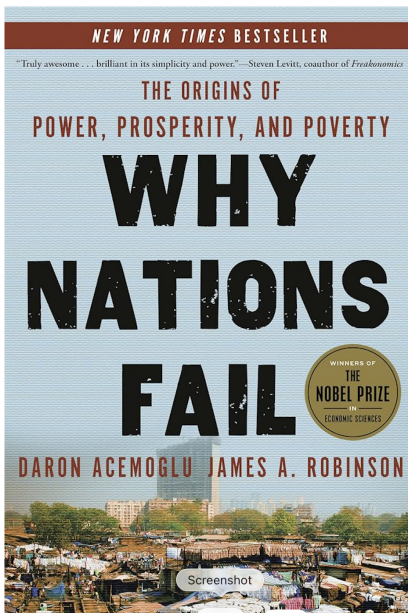
Different social sciences focus on different domains:

- **Economics** → Economic behavior (resource distribution. . .)
- **Political Science** → Political institution and behavior (government, politics. . .)
- **Sociology** → Social behavior (social structure, culture. . .)
- **Psychology** → Human actions, reactions, and mental processes
- . . .

Astronomers vs. Ornithologists vs. Political Scientists



Why Don't Political Scientists Prescribe?



Why Don't Political Scientists Prescribe?

- Practitioners know the practice better than we do.
- They may simply not listen. Their priorities—personal wealth, power, and political survival—might conflict with broader social welfare.



Kim Jong Un and His Luxury Car



Ordinary North Koreans

Then, What do Political Scientists do?

We observe and describe.



Political scientists cannot directly change and save the world ,
but we can provide evidence that changes **how people think about politics**
— changing how people think is the first step toward changing the world.

- Challenge conventional wisdom
- Identify causes and consequences
- Reveal patterns and regularities that are otherwise difficult to see
- Inform public debate and policy decisions

What Kind of Questions Do Political Scientists Ask?

Political scientists seek to **describe and explain political phenomena**.

To do so, we primarily ask **positive questions** about *what is, why, how, and when*:

- Why do some people vote while others do not?
- Why do some countries become democratic?
- What explains civil wars?
- What causes international conflict?

These differ from **normative questions**, which prescribe *what ought to be*.

Normative vs. Positive Research

	Normative Research	Positive Research
Question	What ought to be?	What is , why , and how ?
Subjectivity	Involves value judgments	(Ideally) free from value judgments
Falsifiability	Cannot be tested	Can be tested
Policy Implications	Prescriptive	Descriptive

Which of the following statements is an example of a positive political science statement?

- A. The government should adopt stricter gun control laws.
- B. Democracies are better than authoritarian regimes.
- C. Countries with compulsory voting have higher voter turnout.
- D. The United States should increase foreign aid to developing countries.

In-Class Exercise (1)

Which of the following statements is an example of a positive political science statement?

- A. The government should adopt stricter gun control laws.
- B. Democracies are better than authoritarian regimes.
- C. Countries with compulsory voting have higher voter turnout.
- D. The United States should increase foreign aid to developing countries.

Answer: C

Suppose a political scientist states, “Increasing campaign spending will lead to a higher share of the vote.” What type of statement is this political scientist making?

- A. Positive statement
- B. Normative statement
- C. Mixed statement
- D. Neutral statement

In-Class Exercise (2)

Suppose a political scientist states, “Increasing campaign spending will lead to a higher share of the vote.” What type of statement is this political scientist making?

- A. Positive statement
- B. Normative statement
- C. Mixed statement
- D. Neutral statement

Answer: A

In-Class Exercise (3)

Which of the following statements is a normative political science statement?

- A. Voter turnout in the 2024 election was higher than in the 2020 election.
- B. The government should adopt stricter campaign finance regulations.
- C. Countries with proportional representation have higher voter turnout than countries with plurality systems.
- D. Democratic countries are less likely to experience civil war.

In-Class Exercise (3)

Which of the following statements is a normative political science statement?

- A. Voter turnout in the 2024 election was higher than in the 2020 election.
- B. The government should adopt stricter campaign finance regulations.
- C. Countries with proportional representation have higher voter turnout than countries with plurality systems.
- D. Democratic countries are less likely to experience civil war.

Answer: B

Types of Positive Research

- **Theoretical Research**

- Formal Theory
- Verbal Theory

- **Empirical Research**

- **Quantitative Research**: An approach to documenting regularities that relies on **numerical data** (numbers) and statistics.
- Qualitative Research: An approach to documenting regularities that relies on **non-numerical data** (words and images) and news/interviews/anecdotes/archives.

Theoretical and empirical research are integrated.

Quantitative and qualitative research are complementary.

We can turn qualitative data into quantitative data.

This course focuses on quantitative political research.

Rules of The Road to Scientific Knowledge about Politics

Being a political scientist requires:

- Focusing on **causal explanations**.
- Not letting **data alone** drive your theories.
- Basing conclusions on **empirical evidence**.
- Avoiding **normative statements**.
- Pursuing both **generality** and **parsimony**.

Kellstedt & Whitten (3rd ed.), pp. 16–20.

Key Takeaways

- **Normative research** asks what **ought** to be.
- **Positive research** asks what **is**, **why**, and **how**.
- **Normative research** involves value judgments.
- **Positive research** can be evaluated using empirical evidence.
- What types of research questions do political scientists ask?

- This course focuses on **quantitative political research**.

Before our next class:

- Install **R** and **RStudio** on your laptop.
- Bring your laptop to every class.
- An installation guide is available under “**Other Resources**” in the course syllabus.